Muscle Stimulation using PID Control

Closed-loop force control of electrically stimulated muscle: ESP32 stimulator, force sensor, identified model and simulation study

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1. Objective

Hold the force of a stimulated muscle at a reference value, in newtons, while the muscle fatigues and its gain changes.

Structure: SISO. One input, the stimulation intensity u (0 to 1), and one output, the muscle force F (N). Heart rate, skin temperature, pulse current and rail voltage are *safety checks*, not loop inputs.

Mirror mode. The EMG of the healthy opposite limb, normalised to % of maximum voluntary contraction, sets the reference. EMG and force are different quantities and are linked only through this scale.

2. Plant model and controller

Hammerstein model: recruitment nonlinearity, dead time 40 ms, two lags (60 ms, 100 ms), fatigue.

$$r(u) = [\sigma(k(u - u_{50})) - \sigma(-k\,u_{50})] \,/\, [\sigma(k(1 - u_{50})) - \sigma(-k\,u_{50})]$$
$$F = F_{\max} \cdot \varphi \cdot x_2,\ \ F_{\max} = 15\ \text{N},\ k = 9,\ u_{50} = 0.45$$

Controller, 100 Hz, in the force-demand domain:

$$e = F_{\text{ref}} - F_{\text{meas}}$$
$$v = F_{\text{ref}} + K_p\, e + K_i \textstyle\int e\,dt + D,\ \ u = \Phi^{-1}(v)$$

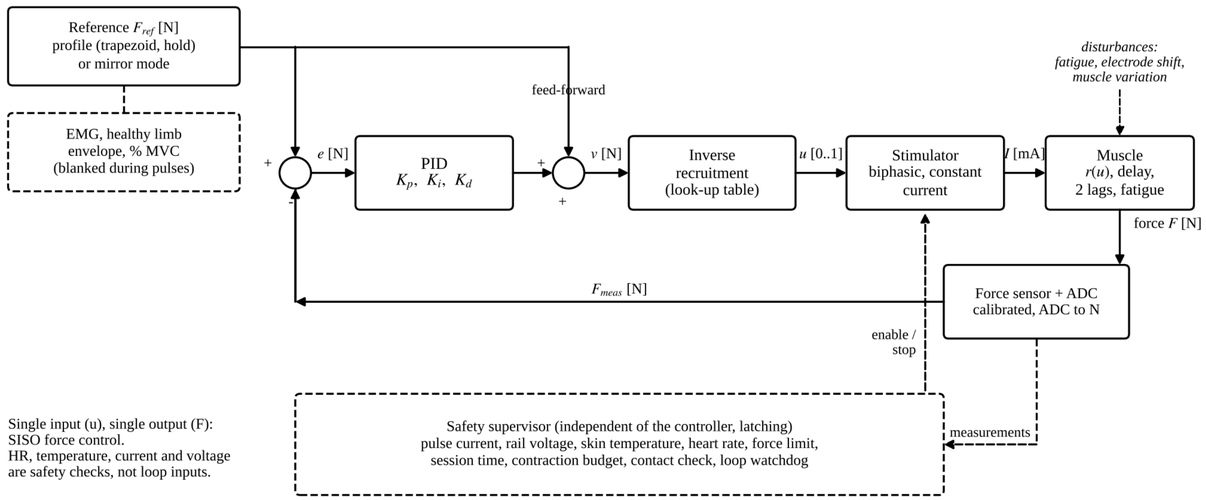
Φ^{-1} is the identified inverse recruitment table. It cancels the S-curve, so the PID sees a plant of gain about 1 N/N. Feed-forward of F_{ref}, back-calculation anti-windup, derivative on the measurement, u slew limit 2 s^{-1}.

Units: F, v [N]; u [–]; I [mA]; K_p [N/N]; K_i [s^{-1}]; K_d [s]; RMSE [N]; IAE [N s].

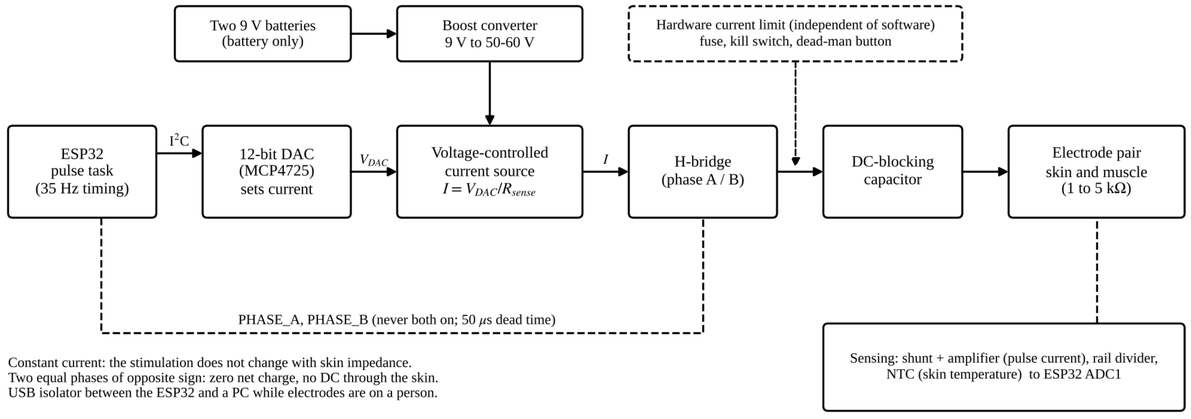
11. Status and limitations

- Model parameters are **assumed**, typical of the literature. The recruitment table shipped in the firmware comes from the simulation, not from a person.
- Firmware: unit-tested on a PC, the C++ controller reproduces the Python one, the sketch compiles against stub headers. **Not run on hardware.**
- The high-voltage analogue stage is specified by function only, no verified schematic.
- No measurements on a person. Simulated and measured results are kept in separate folders.
- PID gives only a small error reduction over PI at about three times the control effort.

3. Control structure



4. Stimulation stage (corrected)



5. Identification and tuning

- **Staircase** (open loop, ascending u): plateau forces give the recruitment curve, stored as a look-up table in the firmware.
- **Step** of force demand 3 → 6 N: fit $K\,e^{-\theta s} / ((\tau_1 s+1)(\tau_2 s+1))$.
- **SIMC rules** with τ_c = effective dead time.

Simulated identification	Value
K	1.11 N/N
τ_1, τ_2	0.074 s, 0.074 s
θ (dead time)	0.056 s
fit R^2	0.997
PI: K_p, K_i	0.54, 4.80 s^{-1}
PID: K_p, K_i, K_d	0.89, 6.0 s^{-1}, 0.033 s

Force sensor: calibrated against known masses, 16-point table, leave-one-out error reported.

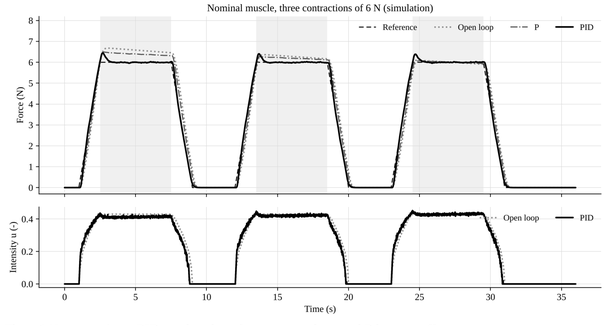
6. Corrections to the first version

First version	Now
“MIMO”	SISO: one input, one output
PWM about 490 Hz, one polarity	Biphasic, charge balanced, 35 Hz, 300 µs per phase, constant current
EMG and FSR compared directly	EMG in % MVC, force in N, explicit scale
FSR as muscle force	Force on a rigid pad, calibrated
ACS712 on an Uno, about 26 mA per ADC step	Shunt, amplifier, 12-bit ADC, hardware limit
Double-integrator muscle	Recruitment, dead time, lags, fatigue
No stimulator hardware	Boost supply, current source, bridge, blocking capacitor, kill switch

12. Next steps

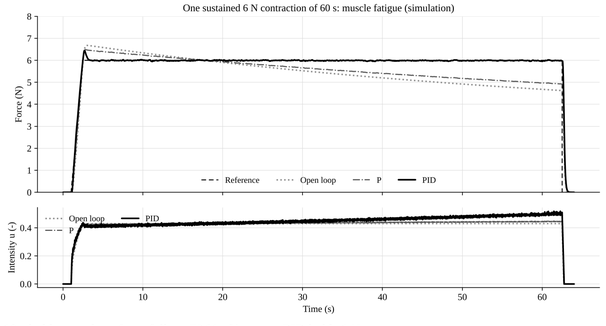
1. Build and verify the stimulator on a dummy load.
2. Calibrate the force sensor, report its leave-one-out error.
3. Measure the recruitment curve and step response, retune.
4. Closed-loop trials with supervision, then mirror mode.

7. Simulation: tracking (simulated)



Three 6 N contractions: PID reaches the reference, open loop and P keep an offset.

8. Simulation: fatigue (simulated)



60 s hold: open-loop force falls to 72 % of its start, PID holds 6 N.

9. Results (5 noise seeds, simulated)

RMSE (N)	OL	P	PI	PID
Nominal	0.45	0.30	0.13	0.12
Gain × 0.7	1.18	0.91	0.24	0.21
Gain × 1.3	1.61	1.02	0.17	0.14
Fatigue: end / start	72 %	79 %	99.9 %	99.9 %

Inverse recruitment cuts PI error 2.3 to 3 times at 3, 6 and 9 N. PID gains little over PI and moves u about 3 times more.

10. Safety supervisor

Latching faults: pulse current > 35 mA, rail > 70 V, skin > 40 °C or +4 °C, heart rate outside 40 to 150 min^{-1}, force > 10 N, session > 300 s, 15 s continuous budget, no current measured, loop late > 50 ms. Dead-man button, kill switch, hardware current limit.

Stimulation parameter	Value
Waveform	biphasic, symmetric
Rate	35 Hz
Phase width, gap	300 µs, 50 µs
Current at u = 1	30 mA (limit 35 mA)
Duty cycle	about 2 %